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—

Δ

o_H,o_D,o_A

$$p_r=\frac{1/o_r}{\sum_{s\in\{H,D,A\}}1/o_s},r\in\{H,D,A\},$$

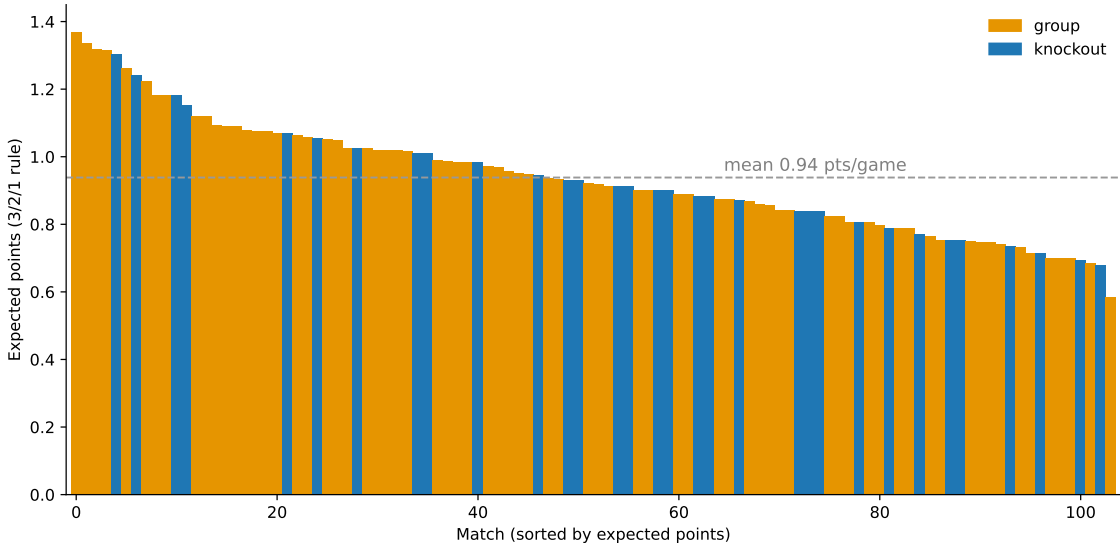
λ_h,λ_a

$$(\hat{\lambda}_h,\hat{\lambda}_a) \, = \, \sum_{\lambda_h,\lambda_a} \sum_{r \in \{H,D,A\}} \left(\pi_r(\lambda_h,\lambda_a) - p_r\right)^2 1.6 \leq \lambda_h + \lambda_a \leq 3.4,$$

$\pi_r(h,a)rg$

$$(h,a) \, = \, P(h,a) \, + \, P(\, = r, \, = g) \, + \, P(\, = r),$$

$(\hat{\lambda}_h,\hat{\lambda}_a)$



$$E_{AB} = \left(1 + 10^{-\Delta/400}\right)^{-1}, \Delta = R_A - R_B,$$

$$p_D = 0.30\,e^{-|\Delta|/700} p_H = E_{AB} - p_D/2 R - 20 - 15 - 10 - 10 - 5 + 40 + 20$$

$$N = 200,000 p \sqrt{p(1-p)/N} \leq 0.11$$

$$-N$$

$$s$$

$$T_s^* ~=~ {}_T P(Ts),$$

$$qp$$

$$D(q\parallel p) ~=~ \sum_t q_t\,2^{\frac{q_t}{p_t}}$$

$$D$$

$$D(q\parallel p)q$$

$$\rightarrow$$

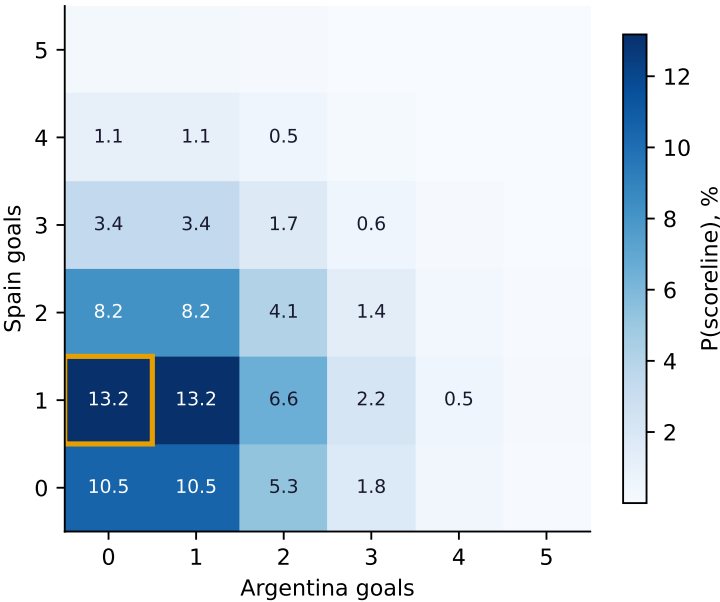
$$p_h+p_d/20.30$$

$\Delta -$

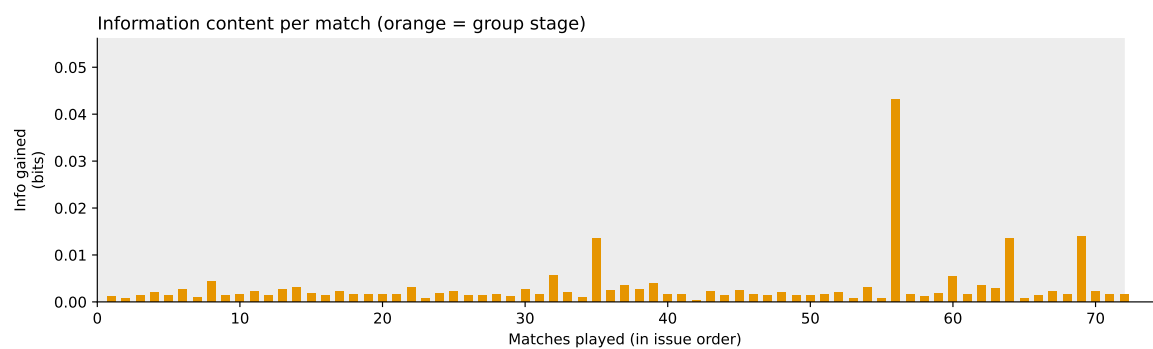
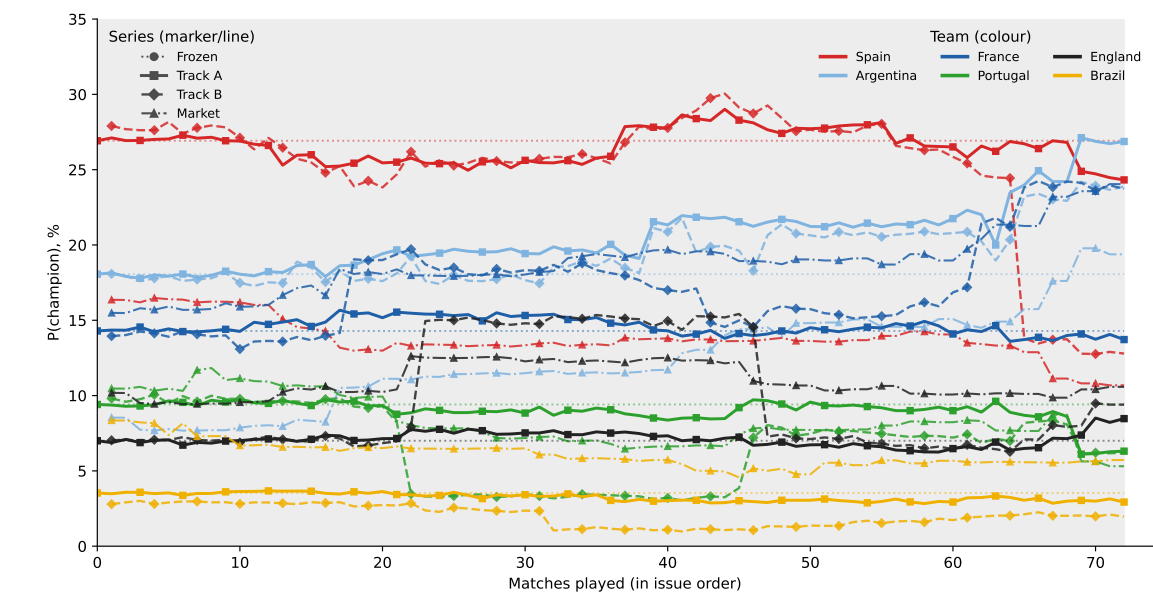
`Ndata/frozen_stage_probs.json` $N\Delta -$

$\rightarrow$
 $\rightarrow$
 $\rightarrow$

+4.0
+2.4
+9.6



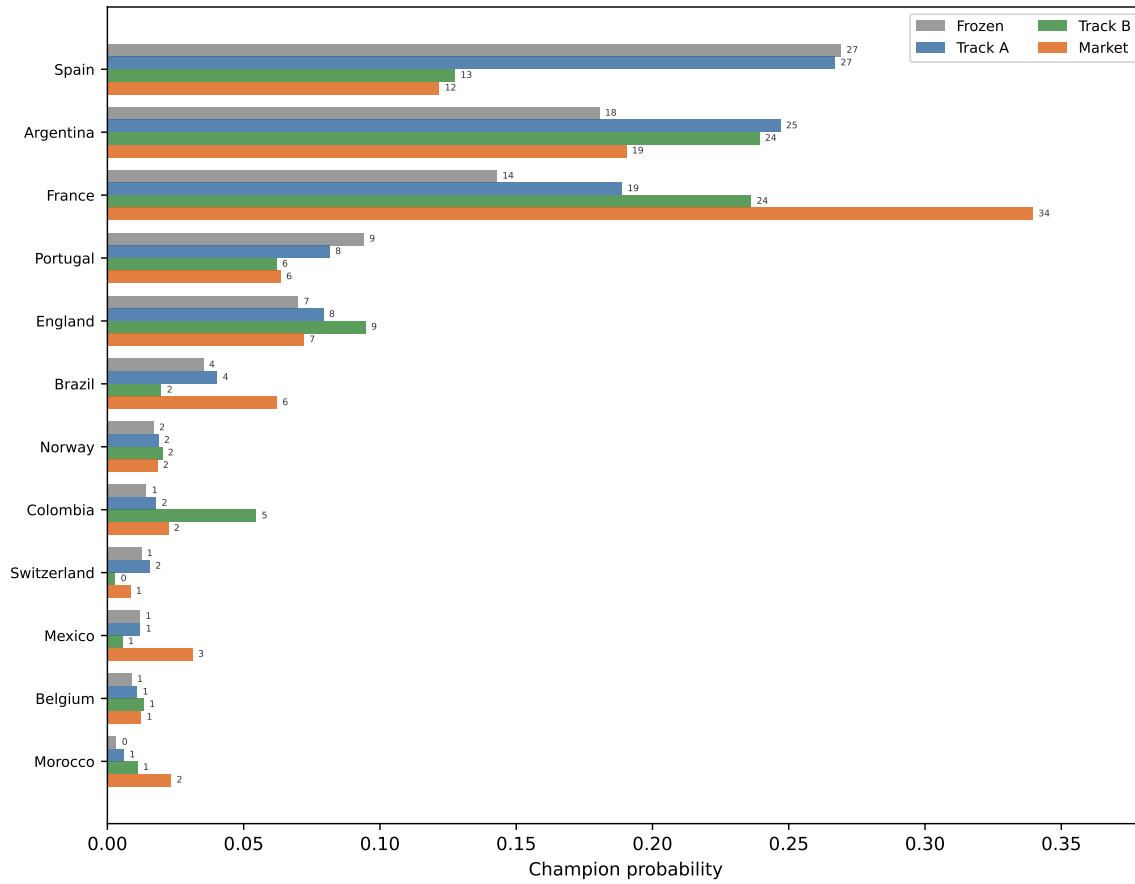
$$\hat{\lambda} = 1.25\hat{\lambda} = 1.00$$



$\Delta-$	$\Delta-$	
-14.2	+1.1	-75.5
+5.9	+1.1	-58.6
+9.3	+1.5	+50.1
-3.2	+2.7	-96.2
+2.5	+1.0	-12.8
-1.5	+1.6	-48.7
+0.3	+10.6	-17.6
+4.0	+6.0	+126.0
-1.0	+6.0	-93.1
-0.6	+8.4	-13.5
+0.5	+3.6	+56.1
+0.8	+11.2	+79.3
-0.3	+34.8	-115.4
+0.7	+11.6	+90.4
+0.3	+17.7	+126.1
+0.3	+54.7	+40.5
	+23.4	+8.8
+0.1	+42.6	+92.9
-0.2	+23.9	-56.8
	+37.6	-0.1
	+65.3	+24.1
	+71.0	+57.8
-1.5	+4.6	-99.6
	-7.0	-55.2
	-68.3	-86.2
+0.3	+12.3	+23.6
+1.8	+1.2	+54.5
	+77.1	+107.1
	-12.0	+33.5
-0.1	-62.6	-52.6
	-7.7	-48.5
	+18.6	-22.9
-0.6	+24.0	-64.4
	-28.5	+22.4
-1.8	+9.0	-68.5
	-30.9	-12.2
	-35.9	+5.3
	-24.3	+2.9
	-34.0	+17.0

$\Delta-$	$\Delta-$	
	-69.1	-64.1
	+18.6	+16.0
	+54.7	+154.8
	-66.2	-55.1
+0.8	+35.8	+161.0
	-40.2	-28.2
	-80.4	-51.2
	-87.8	+34.4
	-38.5	-87.6

$> 0.50 < 25$
 $\lambda = 1.20 \lambda = 1.05$
 $\lambda = 1.75 \lambda = 0.60$
 $\lambda = 0.85 \lambda = 1.60$
 $\lambda = 1.65 \lambda = 0.80$
 $\lambda = 0.10 \lambda = 2.00$
 $\lambda = 0.45 \lambda = 1.85$
 $\lambda = 1.25 \lambda = 1.00$
 $\lambda = 0.20 \lambda = 1.80$
 $\lambda = 2.20 \lambda = 0.10$
 $\lambda = 1.75 \lambda = 0.70$
 $\lambda = 1.70 \lambda = 0.75$
 $\lambda = 1.85 \lambda = 0.10$
 $\lambda = 1.60 \lambda = 0.80$
 $\lambda = 1.70 \lambda = 0.70$
 $\lambda = 2.05 \lambda = 0.10$
 $\lambda = 1.85 \lambda = 0.20$
 $\lambda = 1.60 \lambda = 0.85$



$$\lambda = 1.05 \lambda = 1.15$$

$$\lambda = 2.15 \lambda = 0.10$$

$$\lambda = 0.80 \lambda = 1.60$$

$$\lambda = 1.00 \lambda = 1.40$$

$$\lambda = 1.20 \lambda = 1.05$$

$$\lambda = 1.65 \lambda = 0.80$$

$$\lambda = 1.05 \lambda = 1.05$$

$$\lambda = 1.45 \lambda = 0.95$$

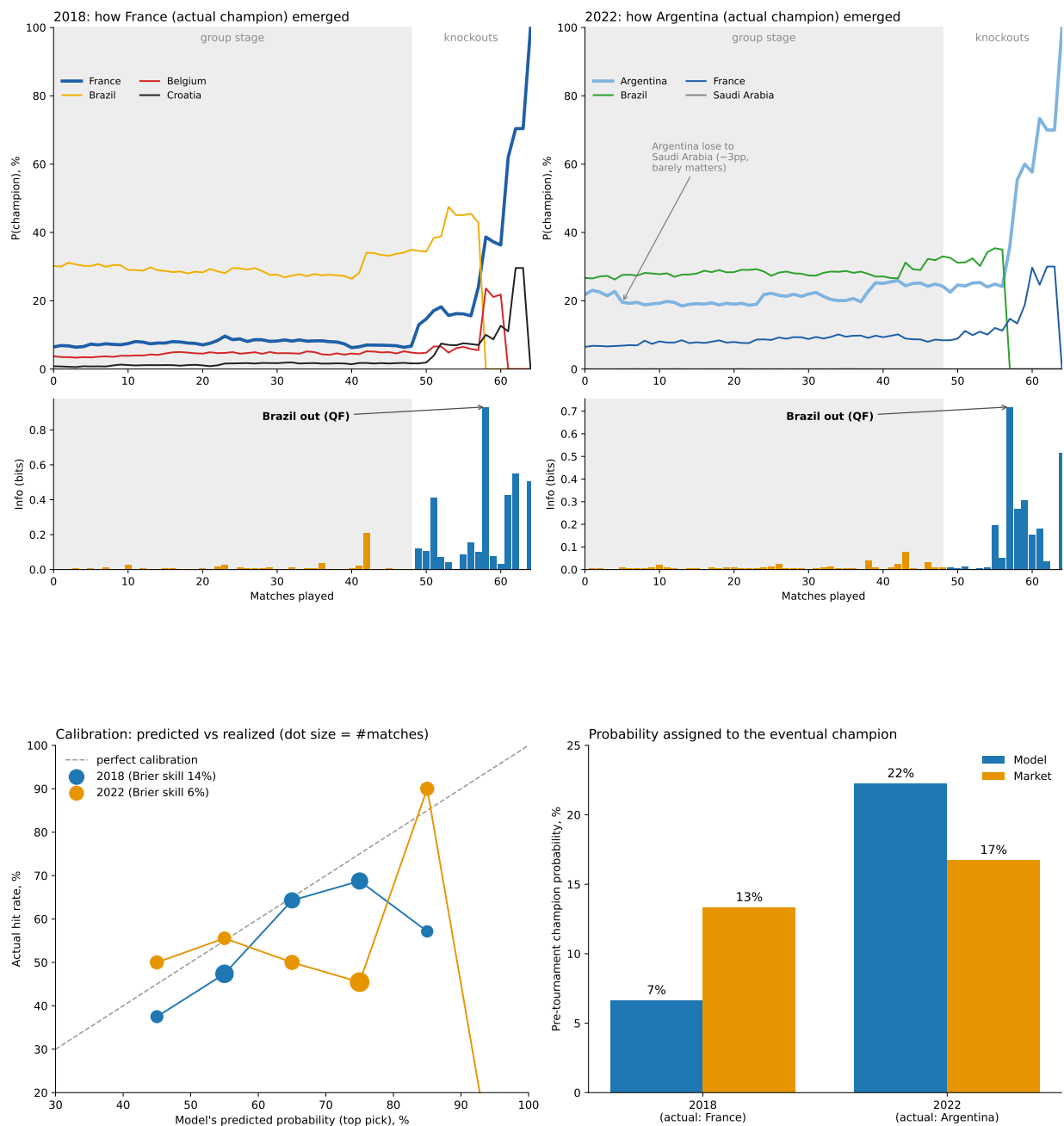
$$\lambda = 1.60 \lambda = 0.85$$

$$\lambda = 1.05 \lambda = 1.05$$

$$\lambda = 1.00 \lambda = 1.35$$

$$\lambda = 1.00 \lambda = 1.25$$

$$p_o q_o \sum_o p_o D(q_o \| q) q q = \sum_o p_o q_o$$

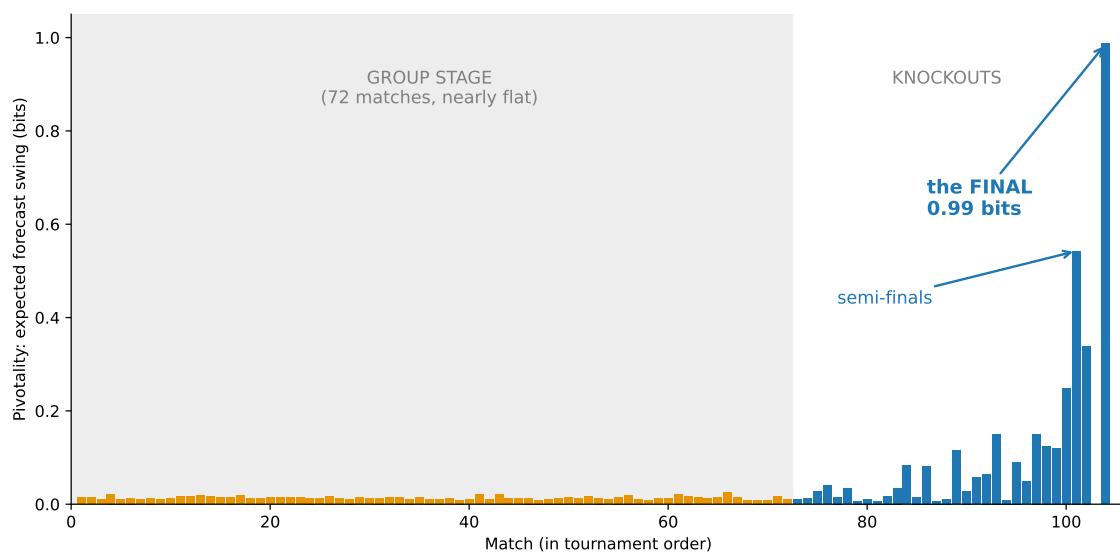
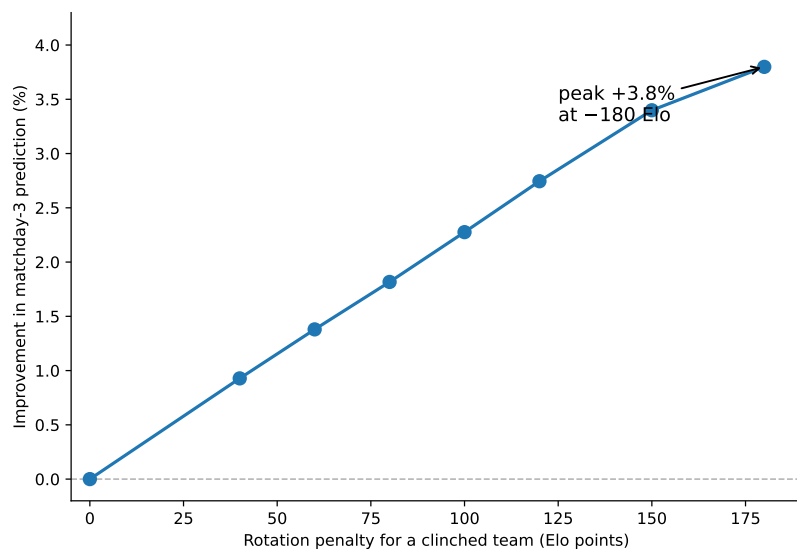


-5.3

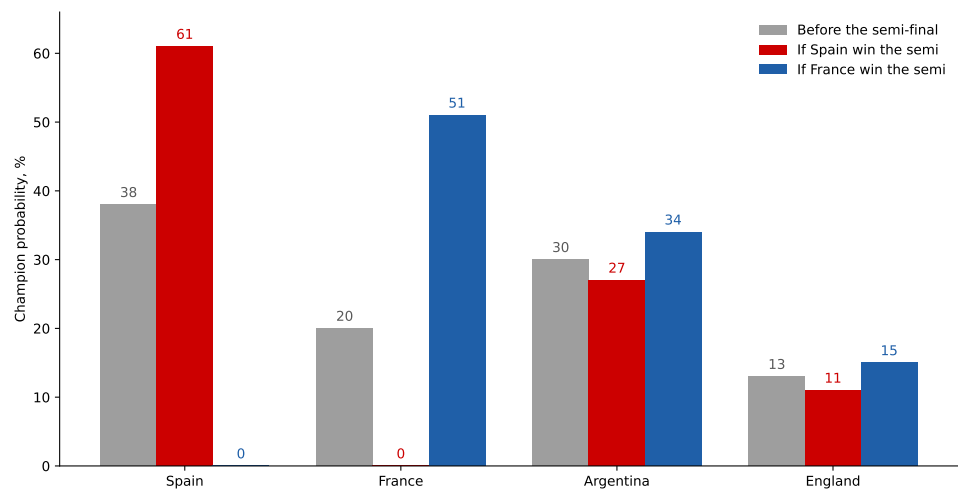
-0.09

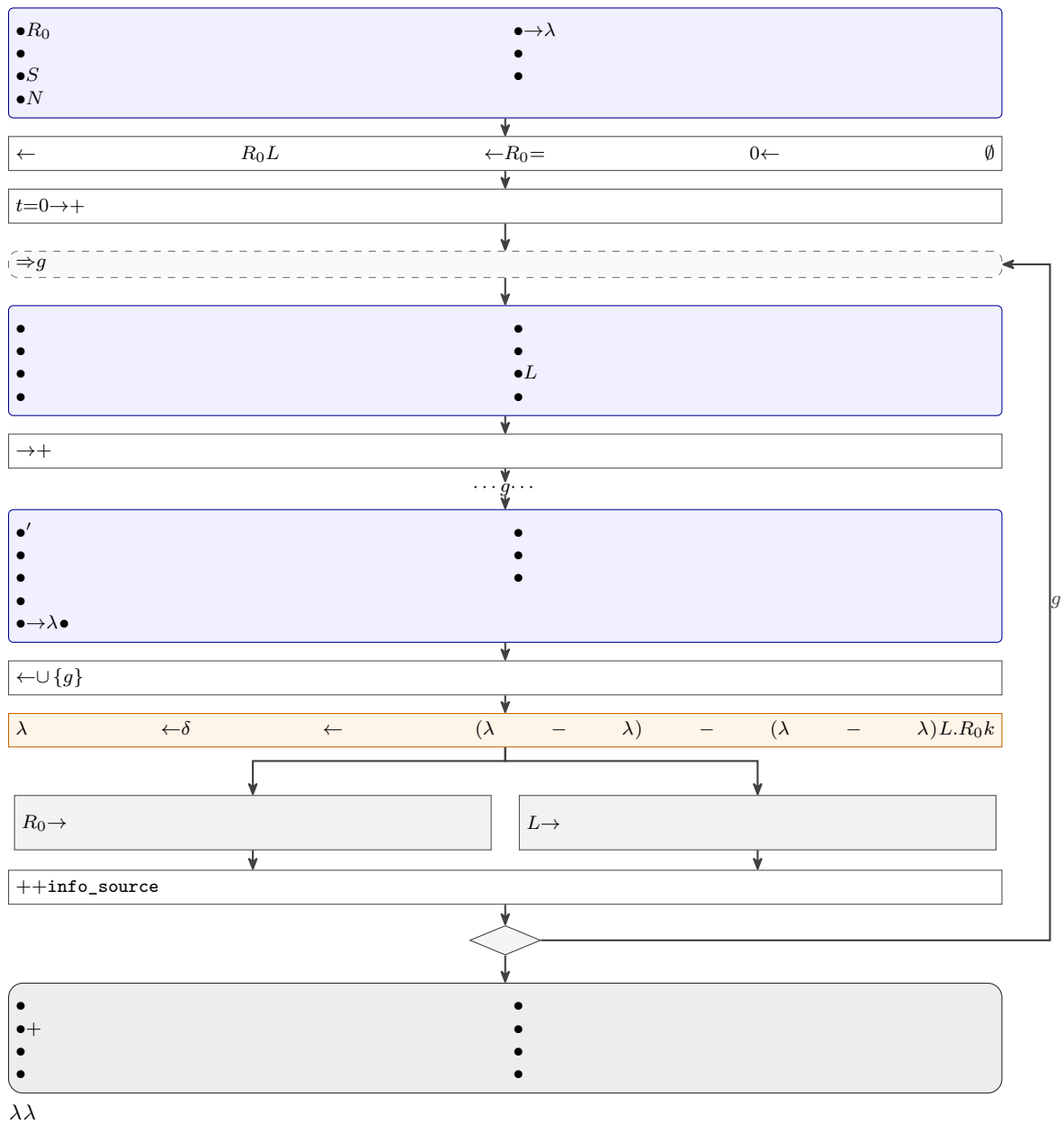
λ

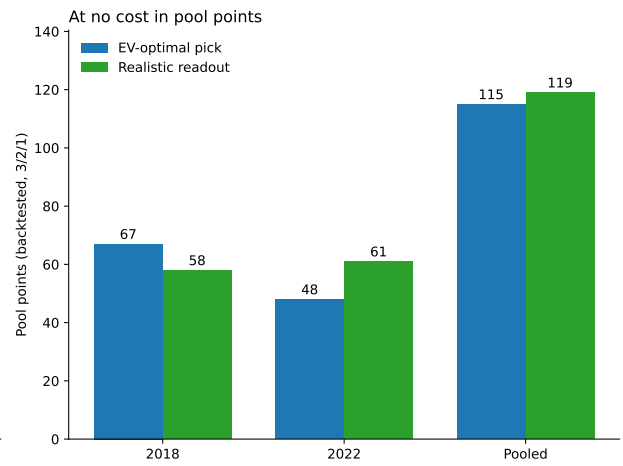
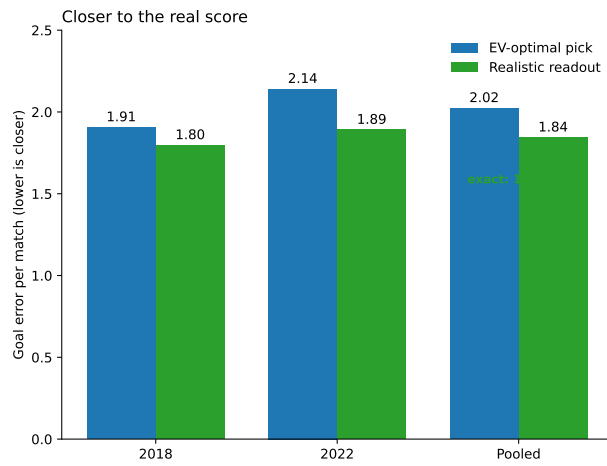
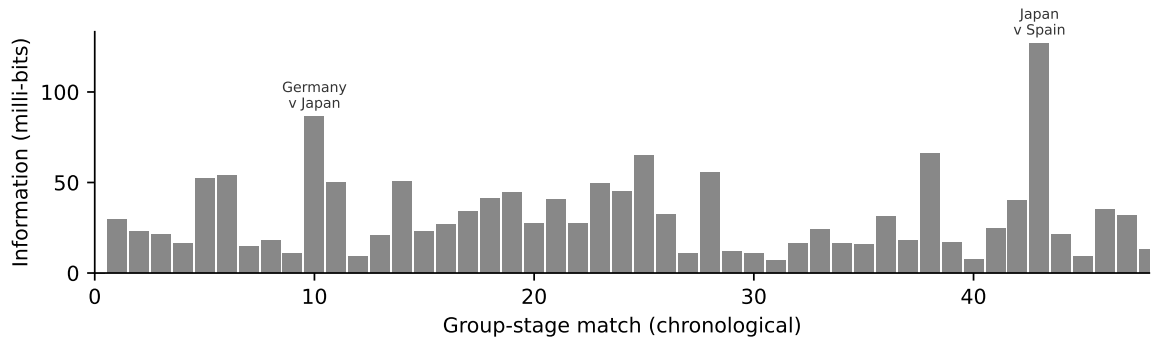
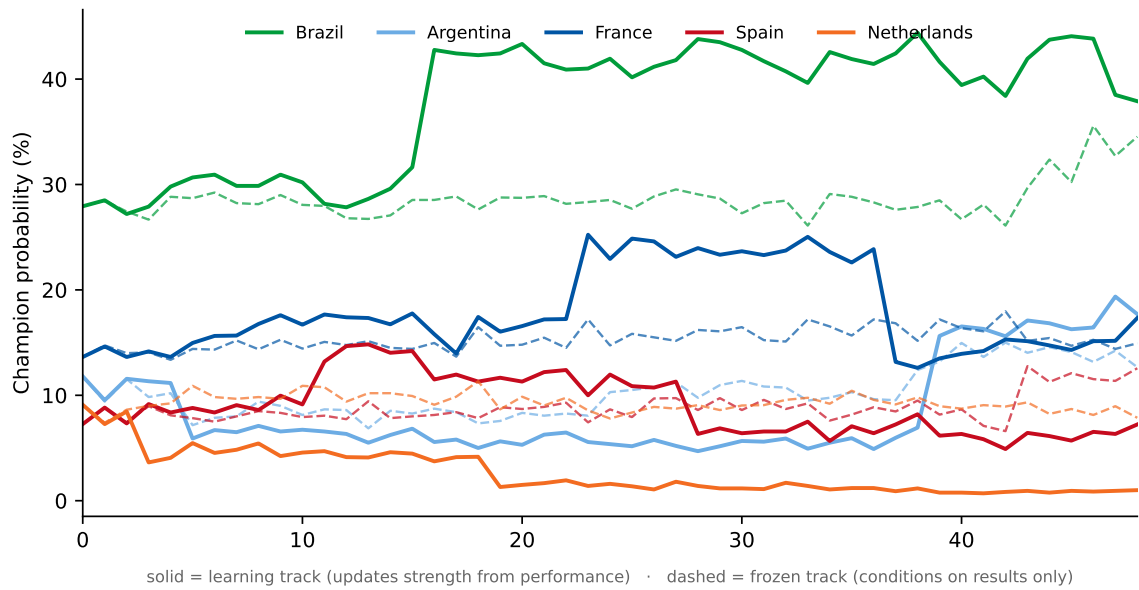
$\rightarrow$

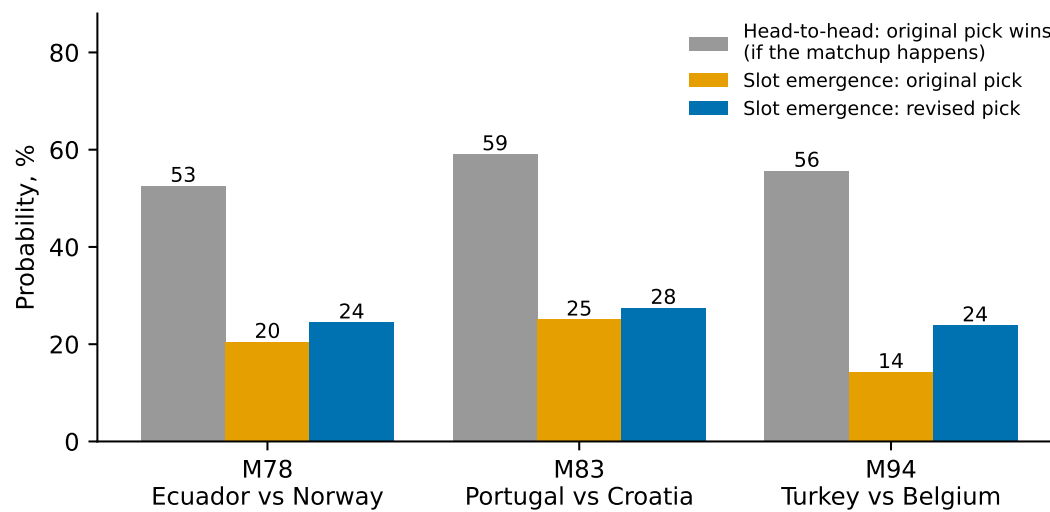


{0.05, 0.10, 0.15}







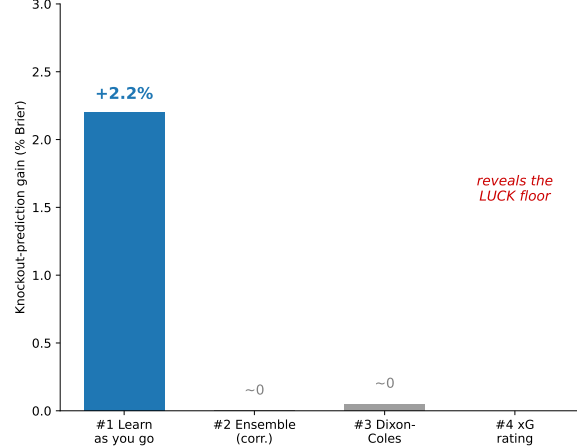


-20-15

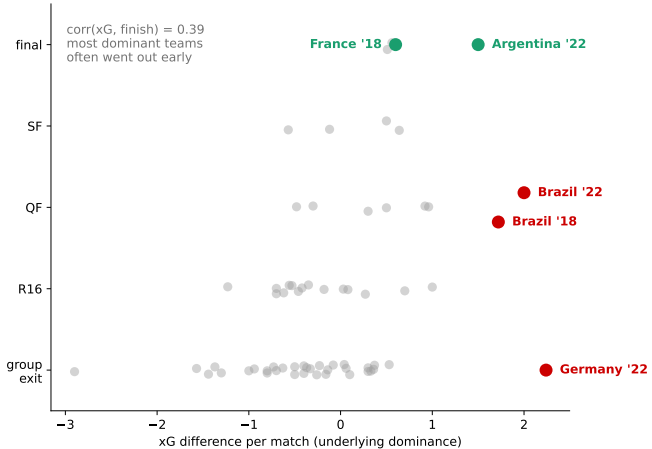
$N = 50,000$

$\lambda \in [0.10, 3.50]$ $N = 200,000$ $N = 20,000$

Improvement audit: only learning helps

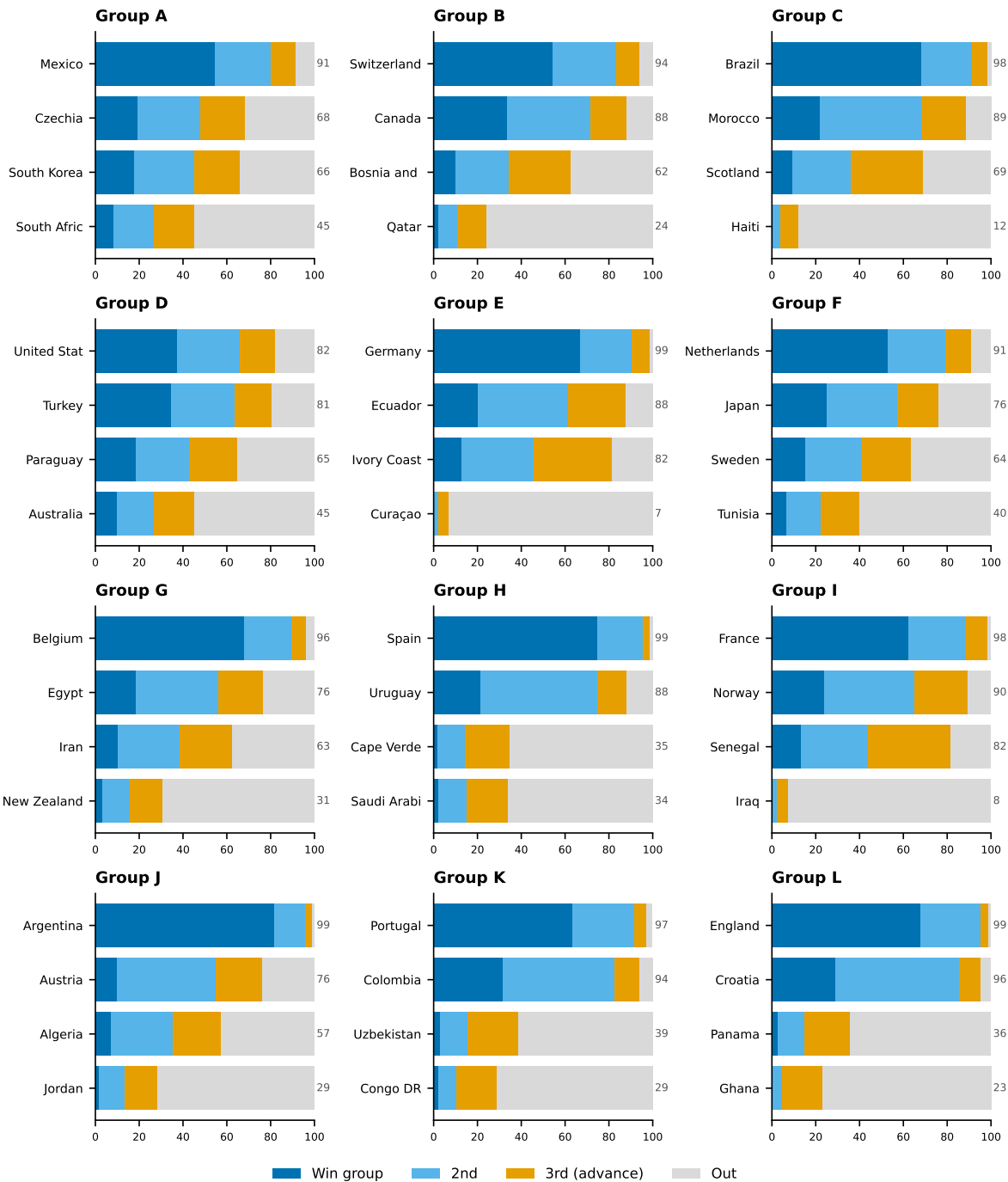


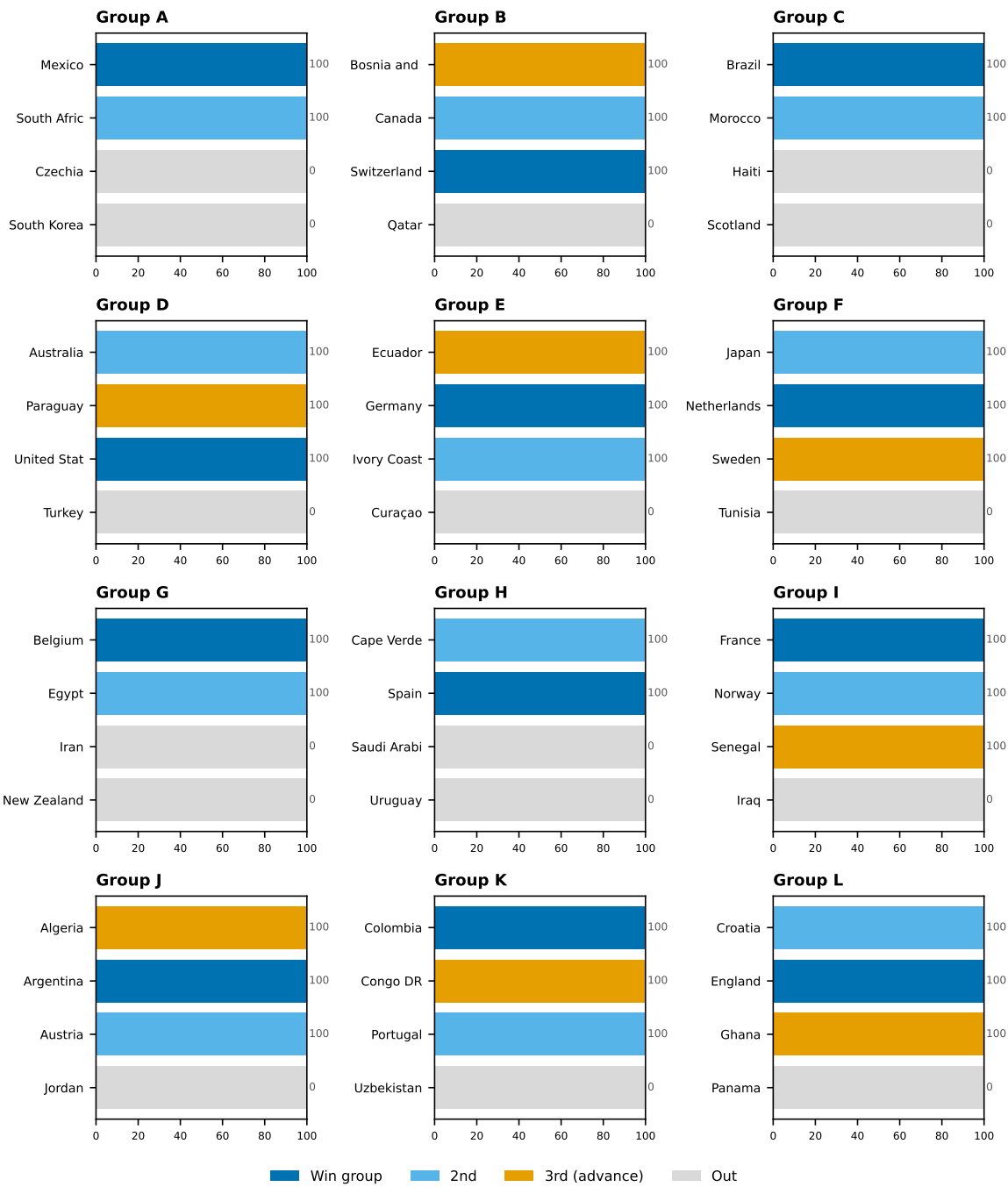
...the rest is luck



-1-2-3

		✓	✓	✓
		✓	×	×
		✓	×	×
✓ ×		✓	×	×
		✓	✓	×
		✓	×	✓
		✓	×	✓
		×	✓	✓
✓ ×				
		×	×	✓
		×	✓	✓
		✓	✓	✓
		✓	✓	✓
✓ ×				





	✓	✓	✓	×
	✓	×	×	×
	✓	×	×	×
	×	×	×	×

✓ ×

✓ ×					
		×	×	✓	✓
		✓	×	×	✓
		✓	✓	✓	
✓ ×					
		×	×	×	✓
		×	×	×	✓
		✓	✓	×	×
✓ ×					
		×	×	✓	✓
		✓	×	×	×
		✓	×	×	
✓ ×					
		✓	✓	✓	✓
		✓	✓	✓	✓
		✓	✓	✓	
✓ ×					
		✓	✓	✓	✓
		✓	×	×	✓
		✓	✓	✓	
✓ × -10					
		×	×	✓	✓
		✓	×	×	✓
		✓	✓	×	×
✓ ×					
		✓	✓	×	✓
		×	✓	✓	✓
		✓	✓	✓	

✓ × △

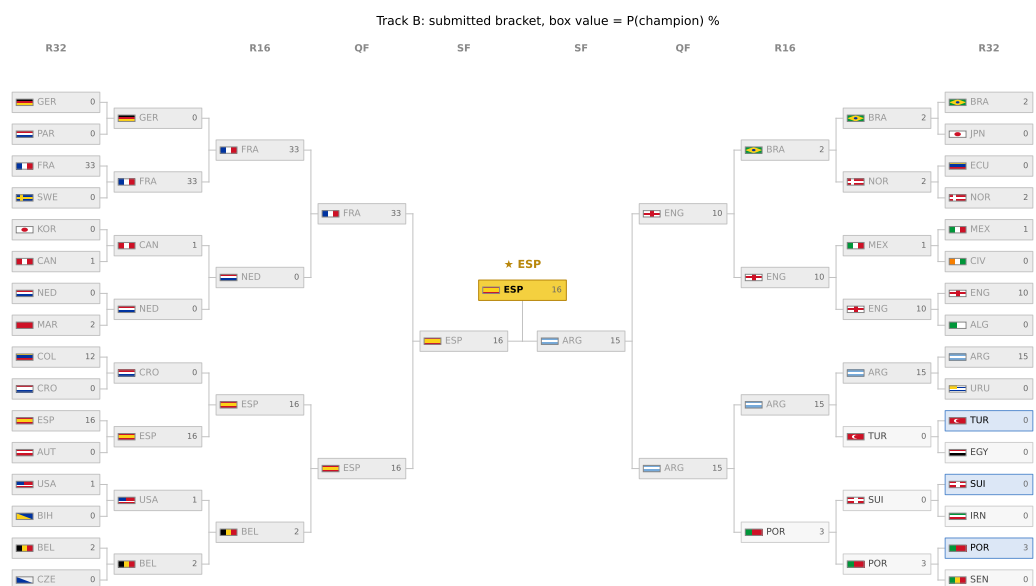
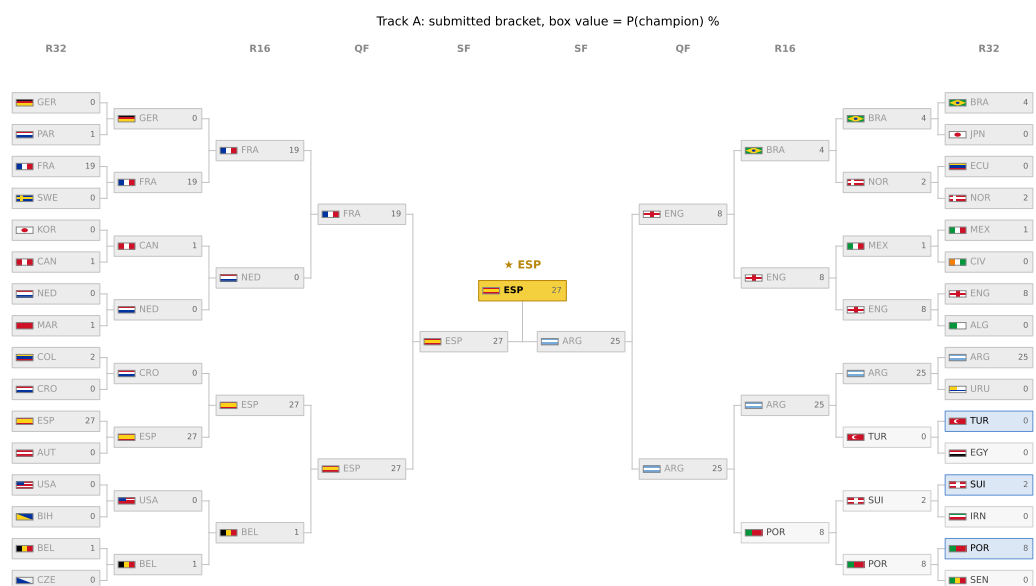
×
✓
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×
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×
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✓

* * *

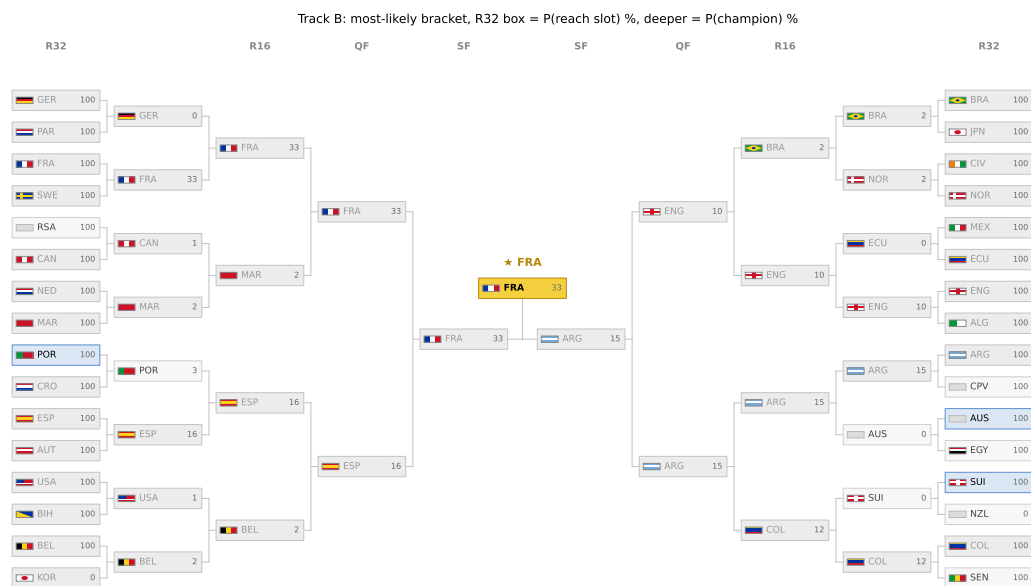
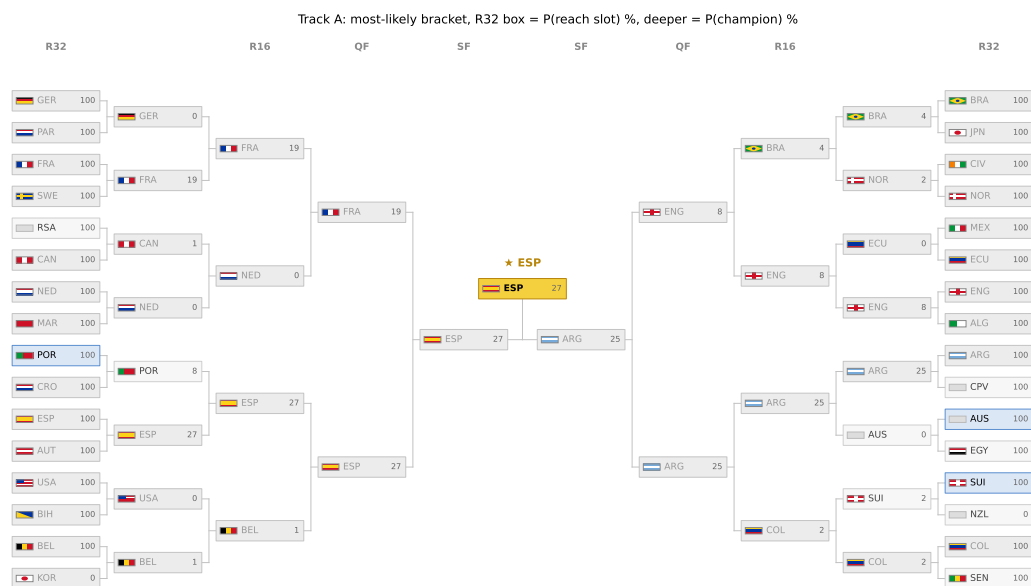
**NNN*



011233scripts/make_bracket_tikz.py



+4.0+2.4



$$|\Delta| > 3$$

+
+
+
+

$$k$$

$$kN=3,000k=40k=0$$

$$k$$

$$N=3,000k=60$$

	1900 → 1736	
	1900 → 1954	
	λ	
	λ	
	+75±75	
	≈	+5.8+6.6
±50×2		+4.8+6.7
λ×0.8×1.2		+7.5+5.3

$$\lambda$$

$$\lambda \;=\; 0.326 \times (),$$

$$\Delta e_0 = (1 + 10^{-\Delta/400})^{-1} p_D = 0.30\,e^{-|\Delta|/700} p_H = e_0 - p_D/2 p_A = 1 - p_H - p_D(\lambda, \lambda)$$

$$\delta \;=\; (\lambda - \lambda) \;-\; (\lambda - \lambda),$$

$$\leftarrow \rho \;+\; (k\,\delta,\,-b,\,+b), \;=\; +\,,$$

$$\rho=0.95b=75kk=0k$$

$$t=0$$

$$kkk=401.662.077.210.619.937.5k=60k4060k=50k=50\rho=0.95b=75k$$

$$kkk=60+7.22.011.76k=40+1.82.692.41$$

$$28.5-8015.30.075023.00.38.918.520892161$$

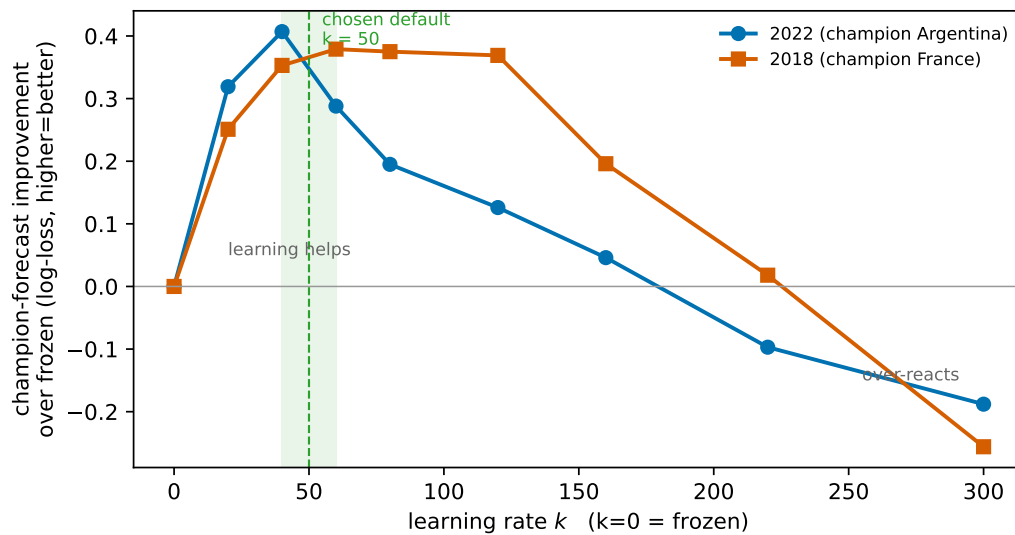
$$\pm50\pm204.87.5$$

$$+6.8-5.3\pm6.7+2.5+0.6\pm4.70.3260.3140.991.091.181.32$$

$$\text{collect_matchperformancefetch_xgcalibrate_proxy}\lambda\text{learnsim_tournamentsnapshotreplay}k$$

$$\text{tournament_2018/2022run_2022_replayrun_2018_ksweepsweep_kpilotstress_test}k=500.95$$

$$75$$



$k=50$

<https://doi.org/10.1086/677350>

<https://doi.org/10.1198/016214506000001437>

<https://doi.org/10.1109/TSSC.1966.300074>

<https://doi.org/10.1111/rssa.12042>

<https://doi.org/10.1515/1559-0410.1418>

<https://doi.org/10.1007/s10994-018-5703-7>

<https://doi.org/10.1287/opre.1070.0448>

<https://doi.org/10.1111/1467-9876.00065>

<https://doi.org/10.1515/jqas-2018-0060>

<https://doi.org/10.1515/jqas-2014-0051>

<https://doi.org/10.1016/j.ejor.2014.10.054>

<https://doi.org/10.1111/1467-9884.00243>

<https://doi.org/10.1080/00031305.1997.10473945>

<https://doi.org/10.1016/j.ijforecast.2009.10.002>

<https://doi.org/10.1287/mnsc.47.3.369.9769>

<https://doi.org/10.1111/1467-9884.00366>

<https://doi.org/10.1016/j.ijforecast.2009.10.001>

<https://doi.org/10.1111/j.1467-9574.1982.tb00782.x>

<https://doi.org/10.1016/j.ijforecast.2014.02.008>